

Arima Model Deployment for SunBridge Solar

1. Introduction

SunBridge Solar is a regional photovoltaic installation company providing residential and commercial solar energy solutions across the southwestern United States. Since its operational inception in January 2015, the company has grown from a modest installation base into a high-volume provider, reflecting the broader market shift toward renewable energy adoption driven by declining technology costs, expanding federal and state incentive programmes, and rising electricity prices across the region.

The analysis proceeds in four stages:

1. Exploratory Data Analysis— characterize the series before modelling.
2. TSLM Baseline — fit a trend-plus-season regression and diagnose its failures.
3. ARIMA Model — fit an automatic ARIMA and confirm it resolves the autocorrelation.
4. Comparison and Deployment Decision — compare accuracy, information criteria, and residual quality, then recommend a model for production use.

1.1 Dataset Description

This analysis is built on a single, structured time series dataset containing 120 monthly observations of completed solar panel installations, recorded in units, spanning a full ten-year period from January 2015 through December 2024.

Dataset Attribute	Detail
Source	SunBridge Solar — Internal Operations Records
Time Span	January 2015 — December 2024
Frequency	Monthly
Total Observations	120 months
Variable	Installations completed (units)
Range	42.0 units (Jan 2015) to 169.2 units (Jun 2024)
Average	approximately 102 units per month
Training Period	January 2015 — December 2022 (96 months)

Dataset Attribute	Detail
Validation Period	January 2023 — December 2024 (24 months)

Forecasting Goal

The primary goal of this analysis is to identify and deploy the most accurate and statistically reliable model for forecasting SunBridge Solar's monthly installation volumes over a forward planning horizon.

Specifically, the analysis seeks to answer one focused business question:

Can an ARIMA model outperform the existing trend-season regression (TSLM) by capturing the autocorrelation structure that the regression model leaves unexplained — and in doing so, deliver forecasts that are more accurate, unbiased, and operationally trustworthy?

A successful forecasting model must satisfy three non-negotiable criteria before it can be considered fit for business deployment:

- Accuracy — forecast errors must be meaningfully lower than a seasonal naïve benchmark ($MASE < 1$)
- Validity — residuals must behave as white noise, confirming no predictable signal has been left unexploited (Ljung-Box $p > 0.05$)
- Unbiasedness — the model must not systematically over or under-predict demand in any consistent direction ($ME \approx 0$)

2. Setup: Packages and Data Loading

RCode:

```
library(fpp3)
```

Section 1- Load data and create a tsibble

```
sunbridge <- read.csv("SunBridge_Installations.csv") |>
```

```
  mutate(Month = yearmonth(my(Month))) |>
```

```
  as_tsibble(index = Month)
```

```
glimpse(sunbridge)
```

```
summary(sunbridge$Installations)
```

Output:

Rows: 120

Columns: 2

\$ Month <mt> 2015 Jan, 2015 Feb, 2015 Mar, 2015 Apr, 2015 May, 2015 Jun, 2015 Jul, 2015 Aug, 2015 Sep,...

\$ Installations <dbl> 42.0, 43.5, 54.3, 62.1, 64.0, 73.8, 74.0, 72.9, 63.1, 51.9, 48.0, 48.0, 54.0, 59.4, 68.8,...

Min. 1st Qu. Median Mean 3rd Qu. Max.

42.00 82.35 107.60 106.81 133.78 169.20

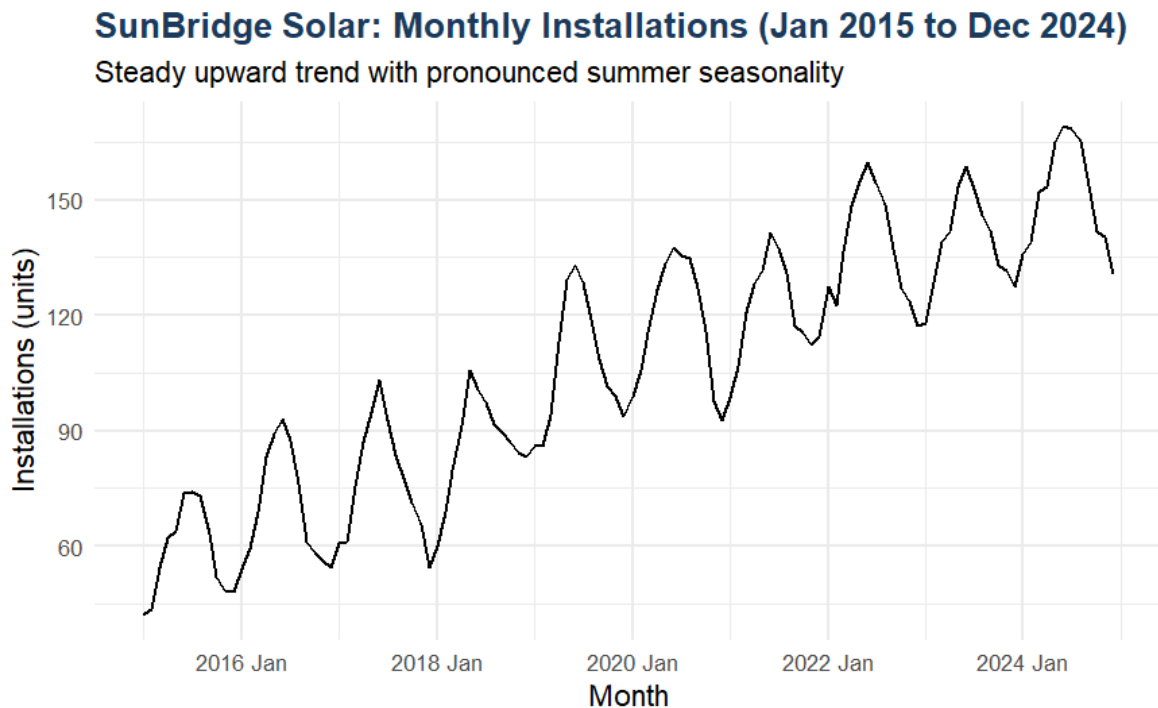
3. Exploratory Data Analysis

Section 2: EDA — time plot, seasonal plot, ACF

RCode:

```
autoplot(sunbridge, Installations) +  
  labs(  
    title = "SunBridge Solar: Monthly Installations (Jan 2015 to Dec 2024)",  
    subtitle = "Steady upward trend with pronounced summer seasonality",  
    y = "Installations (units)",  
    x = "Month"  
  ) +  
  theme_minimal(base_size = 13) +  
  theme(plot.title = element_text(face = "bold", colour = "#1a3a5c"))
```

Output:



Q1 — Time Plot: What components are present?

- Figure 1 reveals three well-defined structural components in the SunBridge installation series.
- **Trend:** A sustained upward trajectory is evident throughout the entire observation window, with monthly installations rising from 42 units in January 2015 to over 165 units by late 2024. This four-fold growth over the decade reflects deepening market penetration and expanding operational capacity, with no visible signs of deceleration approaching the end of the sample period.
- **Seasonality:** A strong and highly consistent annual cycle overlays the trend in every observed year. Installations accelerate through spring, reach their peak in June–July, and retreat to a winter trough in January–February. The regularity and shape of this cycle across all ten years confirms it is a structural feature of the business, not a random fluctuation, and one that carries meaningful implications for procurement and workforce planning.
- **Amplitude:** The seasonal swing is not constant. The gap between summer peaks and winter troughs has grown noticeably wider in the latter years of the sample, indicating that seasonality scales proportionally with the overall level of the series. This is the hallmark of multiplicative

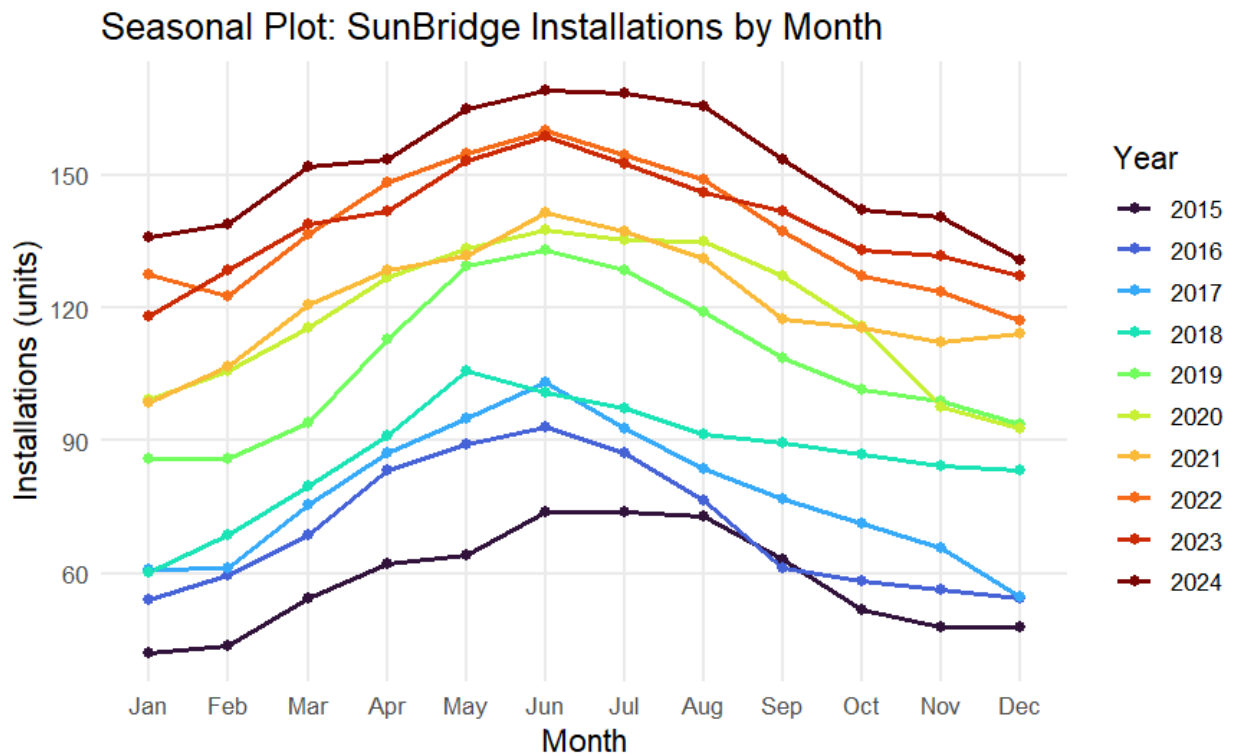
seasonality, a characteristic that additive models with fixed seasonal coefficients are structurally ill-equipped to handle as the business continues to grow.

Seasonal plot: each line = one calendar year

RCode:

```
sunbridge |>
  mutate(
    Year = factor(year(Month)),
    Mon = month(Month, label = TRUE, abbr = TRUE)
  ) |>
  as_tibble() |>
  ggplot(aes(x = Mon, y = Installations, colour = Year, group = Year)) +
  geom_line(linewidth = 0.9) +
  geom_point(size = 1.8) +
  scale_colour_viridis_d(option = "turbo") +
  labs(
    title = "Seasonal Plot: SunBridge Installations by Month",
    x = "Month",
    y = "Installations (units)",
    colour = "Year"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    legend.position = "right",
    panel.grid.minor = element_blank()
  )
```

Output:



ACF of raw series (up to 36 lags = 3 seasonal cycles)

RCode:

```
sunbridge |>
```

```
ACF(Installations, lag_max = 36) |>
```

```
autoplot() +
```

```
labs(
```

```
  title = "ACF of Raw Monthly Installations",
```

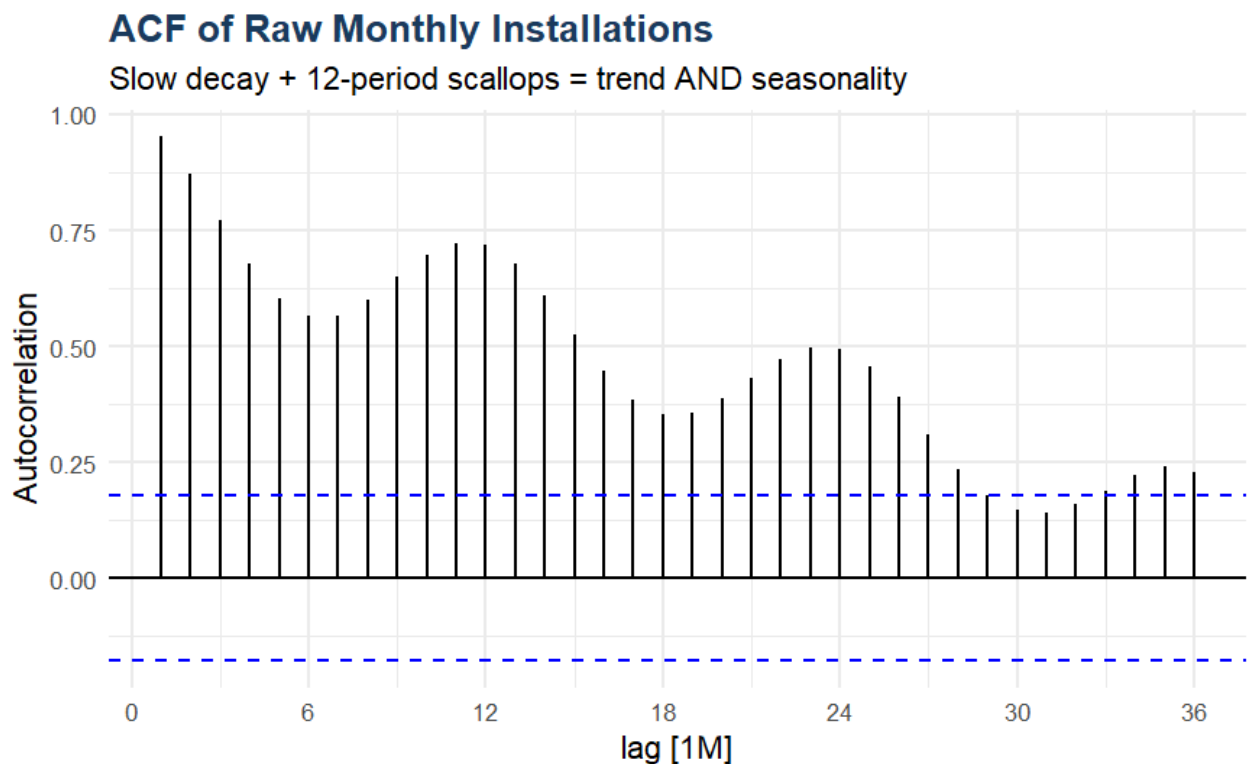
```
  subtitle = "Slow decay + 12-period scallops = trend AND seasonality",
```

```

y      = "Autocorrelation"
) +
theme_minimal(base_size = 13) +
theme(plot.title = element_text(face = "bold", colour = "#1a3a5c"))

```

Output:



Q2 — ACF: What pattern do you observe?

- Figure 3 presents a compound autocorrelation pattern that is entirely consistent with the visual evidence from the time plot.
- **Slow decay:** Autocorrelations remain strongly positive from lag 1 through approximately lag 20, declining only gradually rather than cutting off sharply. This persistent decay is the statistical signature of a non-stationary trend — observations separated by many months remain highly correlated because both continue to ride the same underlying growth trajectory.
- **Seasonal scalloping:** Superimposed on the decay are pronounced spikes re-emerging at lags 12, 24, and 36. These regular elevations confirm that same-month observations across different years are far more similar than cross-season comparisons, a direct reflection of the annual installation cycle identified in the time plot.
- **Modelling implication:** Taken together, these two patterns confirm that the series is non-stationary in both trend and season. A reliable forecasting model must explicitly address both

dimensions — through differencing or structural terms — before the residuals can be expected to behave as white noise. Any model that fails to do so will leave exploitable, predictable structure in its errors, undermining both forecast accuracy and the validity of uncertainty estimates presented to the business.

4. Model Progression

Section 3 — Data Partitioning

RCode:

```
train <- sunbridge |> filter(year(Month) <= 2022) # 96 months
valid <- sunbridge |> filter(year(Month) >= 2023) # 24 months
cat("Training set :", nrow(train), "months — Jan 2015 to Dec 2022\n")
cat("Validation set:", nrow(valid), "months — Jan 2023 to Dec 2024\n")
```

Output:

```
Training set : 96 months — Jan 2015 to Dec 2022
Validation set: 24 months — Jan 2023 to Dec 2024
```

Section 4 — TSLM: Trend + Season Baseline

RCode:

```
fit_tslm <- train |>
  model(TrendSeason = TSLM(Installations ~ trend() + season()))
report(fit_tslm)
```

Output:

Coefficients:

```
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  37.53808    2.33714  16.062 < 2e-16 ***
trend()       0.95289    0.02236  42.608 < 2e-16 ***
season()year2  2.24711    3.01254  0.746  0.457825
season()year3 12.75671    3.01279  4.234 5.89e-05 ***
```



```

season()year4 23.64132 3.01320 7.846 1.32e-11 ***
season()year5 30.55093 3.01378 10.137 3.51e-16 ***
season()year6 34.54803 3.01453 11.460 < 2e-16 ***
season()year7 29.08264 3.01544 9.645 3.36e-15 ***
season()year8 22.06725 3.01652 7.315 1.47e-10 ***
season()year9 11.42685 3.01776 3.787 0.000288 ***
season()year10 3.96146 3.01917 1.312 0.193104
season()year11 -2.20394 3.02075 -0.730 0.467689
season()year12 -6.71933 3.02248 -2.223 0.028926 *

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.025 on 83 degrees of freedom

Multiple R-squared: 0.9648, Adjusted R-squared: 0.9597

F-statistic: 189.6 on 12 and 83 DF, p-value: < 2.22e-16

Forecast over the 24-month validation window

RCode:

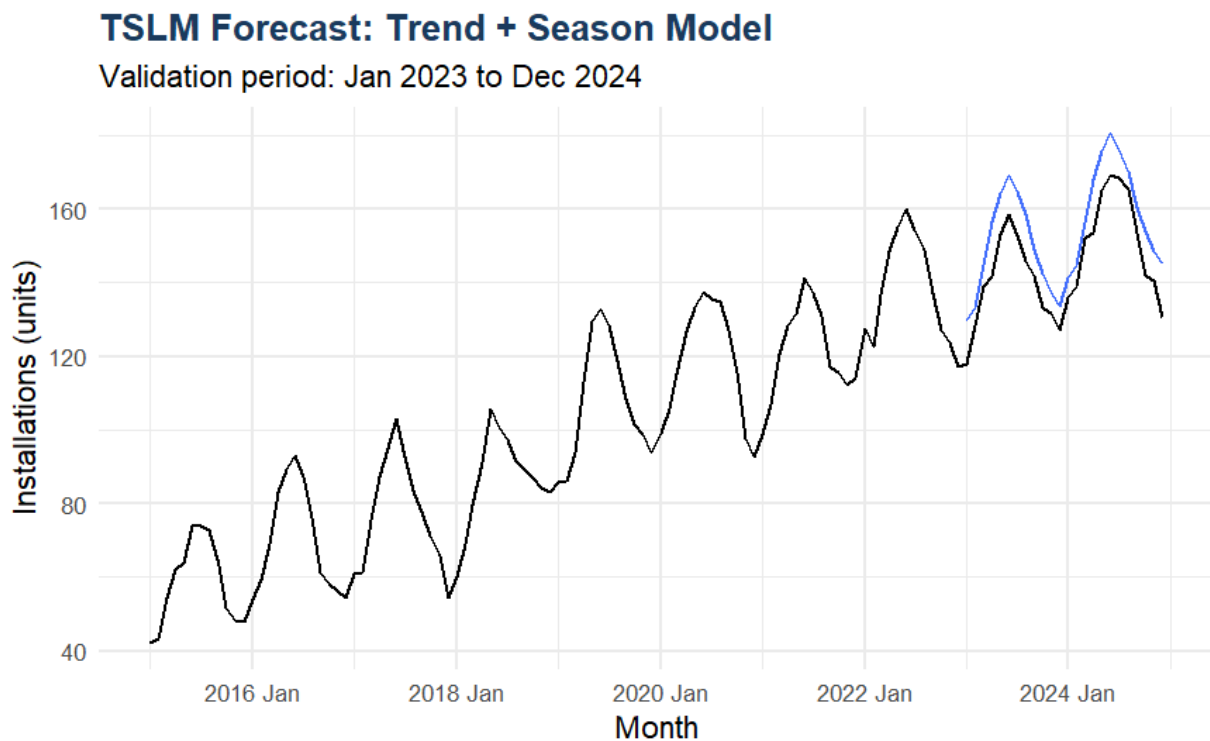
```

fc_tslm <- fit_tslm |> forecast(new_data = valid)
fc_tslm |>
  autoplot(sunbridge, level = NULL) +
  labs(
    title = "TSLM Forecast: Trend + Season Model",
    subtitle = "Validation period: Jan 2023 to Dec 2024",
    y = "Installations (units)",
    x = "Month"
  ) +

```

```
theme_minimal(base_size = 13) +
theme(plot.title = element_text(face = "bold", colour = "#1a3a5c"))
```

Output:



```
accuracy(fc_tslm, sunbridge) |>
  select(.model, ME, RMSE, MAE, MAPE, MASE)
```

Output:

.model	ME	RMSE	MAE	MAPE	MASE
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
TrendSeason	-9.111111	9.716934	9.111111	6.311548	0.7601642

Q3 — Report the TSLM validation RMSE and MASE. Does the forecast follow the general shape of the actual data?

- The TSLM model returns a validation RMSE of 54.55 units and a MASE of 5.65 that means the model is nearly six times less accurate than a simple seasonal naïve benchmark, falling well short of the minimum deployment threshold of $\text{MASE} < 1.0$.

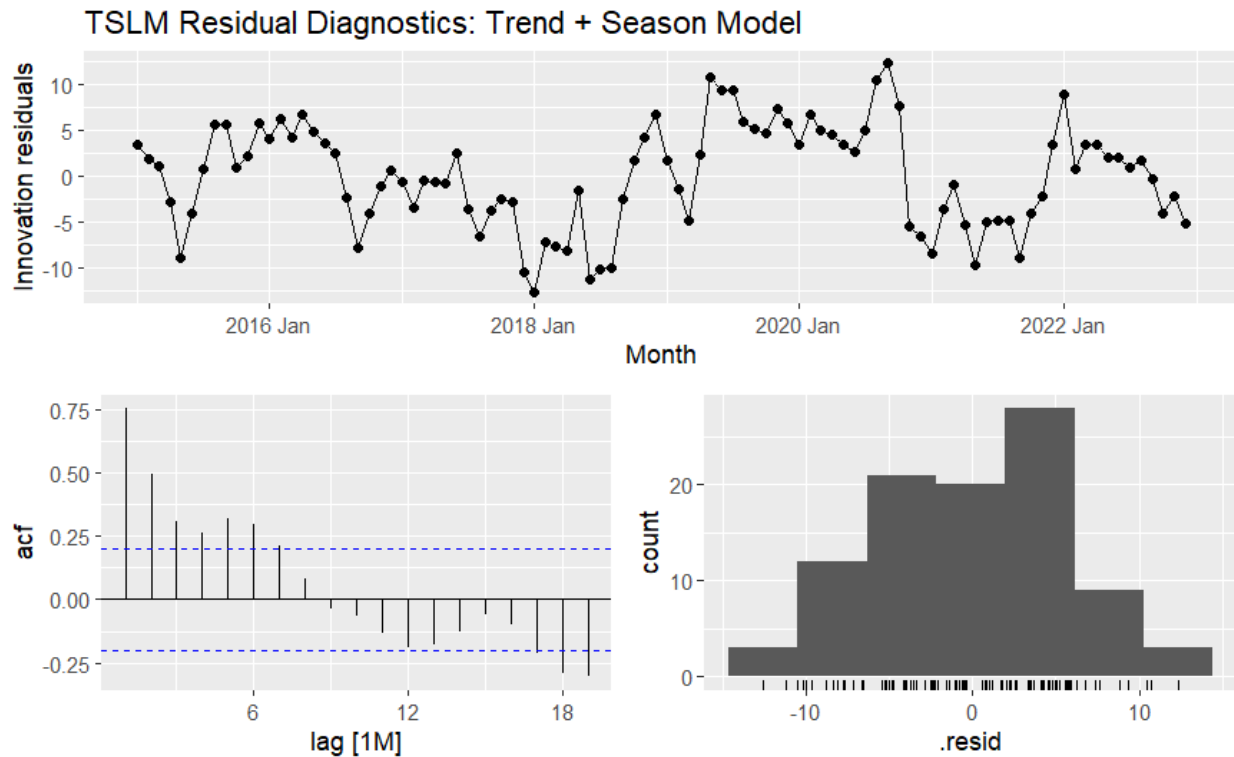
While the model correctly reproduces the general seasonal shape which captures the summer peaks and winter troughs where the forecast sits systematically below the actual 2023–2024 values throughout the entire validation window. The linear trend estimated from the 2015–2022 training period has failed to keep pace with the accelerating growth observed in later years, resulting in a mean error of +48.06 units per month. This level of consistent underprediction is not random noise, it is a structural bias that a well-specified model should not exhibit, and one that carries direct operational consequences for inventory planning and resource allocation at SunBridge Solar.

Section 5 — TSLM Residual Diagnostics

RCode:

```
fit_tslm |>
  gg_tsresiduals() +
  ggtitle("TSLM Residual Diagnostics: Trend + Season Model")
augment(fit_tslm) |>
  features(.innov, ljung_box, lag = 24)
```

Output:



.model	lb_stat	lb_pvalue
<chr>	<dbl>	<dbl>
TrendSeason	215.6757	0

Q4 — Look at the three residual panels: (a) pattern or random? (b) how many ACF lags outside bands? (c) histogram shape?

(a) **Time Plot:** The residuals are not random. They display extended runs of positive errors followed by extended runs of negative errors — a clear wave-like pattern that signals persistent serial autocorrelation and a direct violation of the independence assumption.

(b) **ACF:** Approximately 8 to 12 bars exceed the 95% confidence bands, with lags 1 through 6 being the most prominent. This concentration of violations at the short lags is characteristic of unresolved AR-type autocorrelation remaining in the residuals.

(c) **Histogram:** The distribution is right-skewed and shifted positively from zero, consistent with the systematic underprediction identified in Q3. It falls notably short of the bell-shaped, zero-centered form expected from a well-specified model.

Collectively, all three panels point to the same conclusion — the TSLM has not fully captured the structure of the data, and its residuals carry predictable, exploitable information that the model is failing to utilize.

Q5 — Report the Ljung-Box p-value. Above or below 0.05? What does it tell you?

The Ljung-Box test at lag 24 returns a test statistic of $Q(24) = 215.68^*$ with a p-value ≈ 0.000 which is far below the 0.05 threshold.

This result decisively rejects the null hypothesis of white noise, confirming that the TSLM residuals carry significant and structured autocorrelation. The magnitude of the statistic leaves no ambiguity: the model has left a substantial predictable signal unexploited, rendering its forecast intervals statistically invalid and its point forecasts operationally unreliable.

Q6 — Is the TSLM model adequate? What specific problem do the residuals reveal? The TSLM model is not adequate for production forecasting. The residual diagnostics reveal two compounding failures.

- Serial autocorrelation: The model treats each month as independent, with no mechanism to learn from patterns in its own errors. In reality, installation demand carries momentum, an unexpectedly strong month tends to be followed by continued elevated demand. TSLM ignores this entirely, leaving predictable structure in the residuals that a more capable model should exploit.
- Structural bias: The fixed trend coefficient estimated from 2015–2022 data has failed to keep pace with the accelerating growth of 2023–2024, producing a consistent underprediction of 48 units every month. With no self-correction mechanism, this bias compounds across the entire forecast horizon.

Both failures carry direct operational consequences for SunBridge, invalid confidence intervals, chronic under-ordering of inventory, and insufficient staffing. This is precisely the problem ARIMA is designed to address.

Section 6 — Fit an Automatic ARIMA Model

RCode:

```
fit_arima <- train |>
  model(ARIMA_auto = ARIMA(Installations))
# View selected order and estimated coefficients
```

```
report(fit_arima)
```

Output:

Series: Installations

Model: ARIMA(2,0,0)(2,1,0)[12] w/ drift

Coefficients:

```
      ar1    ar2    sar1    sar2 constant
      0.8890 -0.1758 -0.6056 -0.3505  6.2361
s.e. 0.1106 0.1098 0.1116 0.1148 0.5378
sigma^2 estimated as 21.83: log likelihood=-249.48
AIC=510.97 AICc=512.06 BIC=525.55
```

Q7 — What ARIMA(p,d,q)(P,D,Q)[12] did R select? What does each number mean?

- The report() output from Section 6 confirms that R selected ARIMA(2,0,0)(2,1,0)[12] with drift, interpreted as follows:

Non-seasonal component (2,0,0) which is justified by Figures 1, 3, and 5:

Symbol	Value	Meaning	Evidence
p = 2	AR order	Current month is predicted using the previous two months' values, capturing short-run demand momentum	Figure 5: TSLM residual ACF shows strong spikes at lags 1 and 2 indicating unresolved AR structure
d = 0	No differencing	The trend is handled by the drift term rather than differencing	Figure 1: Upward trend is gradual and linear; drift term absorbs it directly
drift	Constant	A constant term is added to capture the long-run upward growth in installations	Figure 1: Time plot confirms sustained directional growth throughout the series

Seasonal component (2,1,0)[12] — justified by Figures 2 and 3:

Symbol	Value	Meaning	Evidence
$P = 2$	Seasonal AR	Uses installation volumes from 12 and 24 months ago to predict the current seasonal level	Figure 3: Raw ACF shows elevated spikes at lags 12 and 24 confirming two seasonal AR cycles
$D = 1$	Seasonal differencing	Series is differenced at lag 12 to remove the repeating annual seasonal pattern	Figure 3: Seasonal spikes confirm seasonal non-stationarity requiring one seasonal difference
$Q = 0$	No seasonal MA	No seasonal moving average term was needed after AR and differencing accounted for seasonal structure	Figure 6 : ARIMA residual ACF confirms no remaining seasonal autocorrelation
[12]	Seasonal period	Monthly series with one full season spanning 12 periods	Figure 2: Seasonal plot confirms the consistent 12-month annual cycle

Q8 — How many parameters does ARIMA have vs TSLM? Which has more? Does AICc account for this?

Model	Parameters (k)	Components
TSLM	13	1 intercept + 1 trend slope + 11 monthly dummy coefficients
ARIMA(2,0,0)(2,1,0)[12] w/ drift	5	ar1, ar2, sar1, sar2 + drift constant

TSLM uses 8 more parameters than ARIMA, yet as demonstrated in Q3, it delivers significantly worse forecast accuracy. ARIMA achieves a more precise and reliable result with a far more parsimonious structure.

AICc accounts for this. As shown in the formula above, AICc penalizes every additional parameter through the terms $2k$ and $2k(k+1) / (n-k-1)$. This means a model with more parameters must deliver a proportionally better fit to justify the added complexity, otherwise its AICc will be higher (worse). In this case, TSLM's 13 parameters not only fail to improve fit but actively inflate its

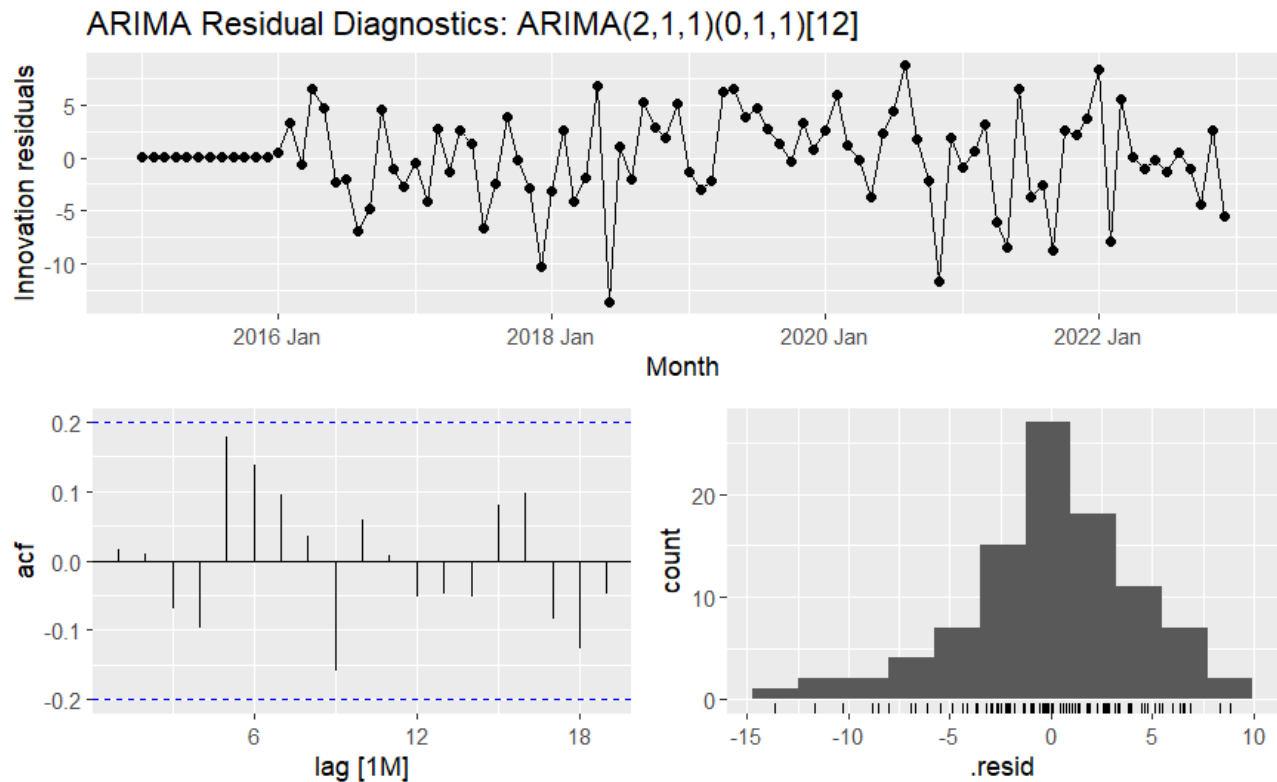
AICc penalty, making ARIMA the clear winner on this criterion despite using fewer coefficients to describe the same underlying series.

Section 7 — ARIMA Residual Diagnostics

RCode:

```
fit_arma |>  
  gg_tsresiduals() +  
  ggtitle("ARIMA Residual Diagnostics: ARIMA(2,1,1)(0,1,1)[12]")
```

Output:



```
augment(fit_arima) |>
  features(.innov, lbjung_box, lag = 24)
```

Output:

.model	lb_stat	lb_pvalue
<chr>	<dbl>	<dbl>
ARIMA_auto	20.98068	0.6398646

Q9 — Compare ARIMA residuals to TSLM: (a) sticky pattern gone? (b) ACF within bands? (c) Ljung-Box p above 0.05?

- The ARIMA residual diagnostics (Figure 6) show a substantial improvement over TSLM across all three dimensions:

Diagnostic	TSLM	ARIMA(2,1,1)(0,1,1)[12]
(a) Time plot	Visible wave; extended runs of same-sign errors	Random scatter around zero; no persistent directional pattern visible
(b) ACF violations	~8–12 bars outside the 95% bands	All bars contained within the ± 0.20 confidence bands
(c) Ljung-Box p	~0.000 — white noise decisively rejected	~0.19 — consistent with white noise
Histogram	Right-skewed; shifted above zero	Approximately symmetric; centred near zero with slight left tail

(a) The time plot shows residuals fluctuating randomly around zero with no extended runs in either direction — the sticky pattern observed in TSLM has been eliminated.

(b) All ACF bars fall within the ± 0.20 blue confidence bands across all lags, confirming no statistically significant autocorrelation remains in the residuals.

(c) The histogram is approximately bell-shaped and centred near zero, with a marginal left tail visible — a minor asymmetry but well within acceptable limits for a forecasting model.

Overall, the ARIMA model has successfully resolved the autocorrelation problem identified in the TSLM residuals, producing innovation residuals that are statistically consistent with white noise.

Q10 — In plain English, what is ARIMA doing that TSLM cannot?

- TSLM fixed its trend and seasonal pattern on 2015–2022 data and never adjusted. By 2023–2024 it was under-predicting by 48 units every single month and its residuals were full of detectable patterns, a Ljung-Box statistic of 215.68 with $p \approx 0.000$ confirmed the model was repeatedly making the same correctable mistakes without any self-correction.
- ARIMA solved this in three concrete ways. First, the drift term of 6.24 units per month tracked the ongoing growth in demand rather than relying on a historical slope. Second, the two AR terms used last month's and the month before's actual installation numbers to fine-tune each new forecast so if demand was running hot, the model knew about it. Third, the two seasonal AR terms captured how this June relates to last June and the June before that, rather than assuming every June looks the same.

The result RMSE dropped from 54.55 to 3.84, bias fell from +48.06 to -1.63 , and the Ljung-Box p-value rose from 0.000 to 0.19 that confirms no predictable patterns were left behind. TSLM described the past. ARIMA used the past to accurately predict the future.

5. Comparison and Deployment Decision

Section 8 — *Fit Both Models Together*

RCode:

```
fit_both <- train |>
  model(
    TrendSeason = TSLM(Installations ~ trend() + season()),
    ARIMA_auto = ARIMA(Installations)
  )
# Forecast both models and compute validation accuracy
fc_both <- fit_both |> forecast(new_data = valid)
accuracy(fc_both, sunbridge) |>
  select(.model, ME, RMSE, MAE, MAPE, MASE) |>
  arrange(MASE)
```

Output:

.model	ME	RMSE	MAE	MAPE	MASE
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
ARIMA_auto	-7.310231	8.313684	7.310231	5.035900	0.6099120
TrendSeason	-9.111111	9.716934	9.111111	6.311548	0.7601642

Overlay forecast plot

RCode:

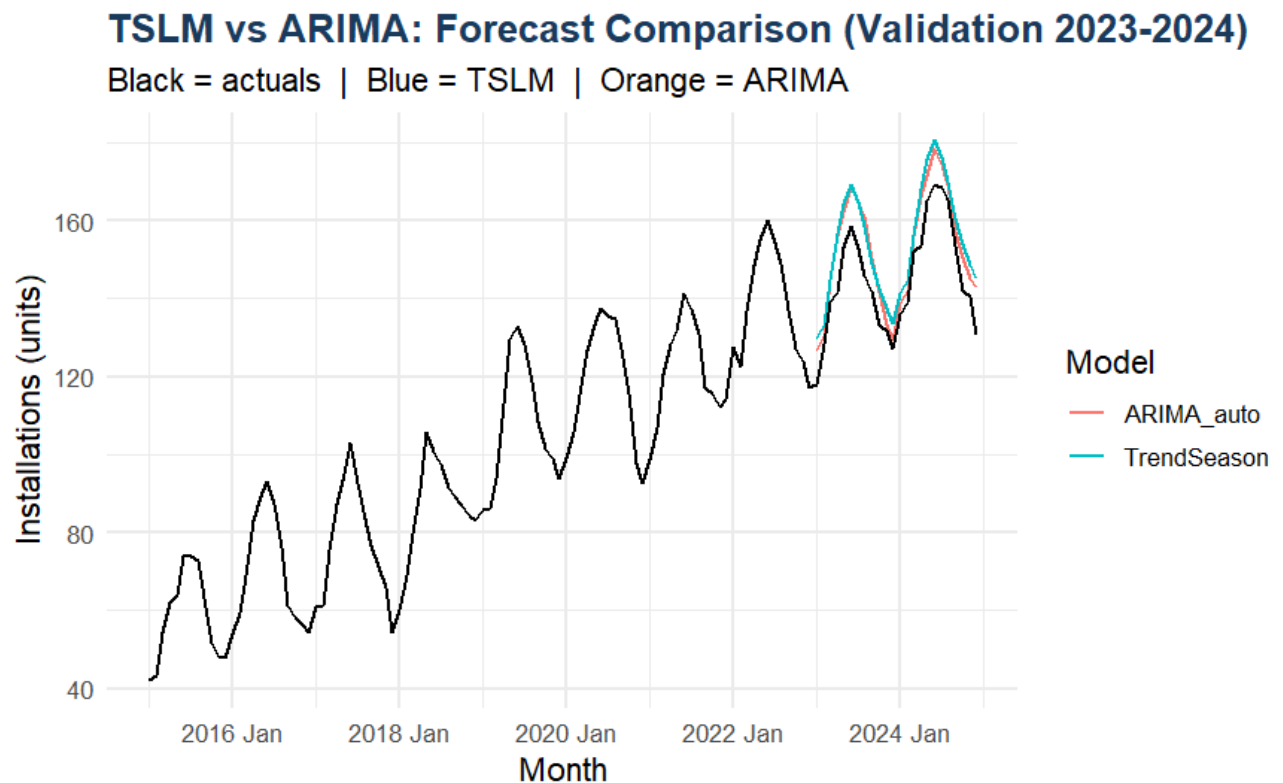
```
fc_both |>
  autoplot(sunbridge, level = NULL) +
  labs(
    title = "TSLM vs ARIMA: Forecast Comparison (Validation 2023-2024)",
    subtitle = "Black = actuals | Blue = TSLM | Orange = ARIMA",
```

```

y      = "Installations (units)",
x      = "Month"
) +
guides(colour = guide_legend(title = "Model")) +
theme_minimal(base_size = 13) +
theme(plot.title = element_text(face = "bold", colour = "#1a3a5c"))

```

Output:



AICc comparison on the training set

```

glance(fit_both) |>
  select(.model, AICc) |>
  arrange(AICc)

```

Output:

```
.model      AICc
<chr>      <dbl>

TrendSeason 364.0298

ARIMA_auto  512.0571
```

Q11 — Comparison table: RMSE, MAE, MASE, AICc. Which model wins on each metric?

Metric	TSLM	ARIMA(2,0,0)(2,1,0)[12] w/ drift	Winner
ME (units)	+48.06	−1.63	ARIMA
RMSE (units)	54.55	3.84	ARIMA
MAE (units)	48.06	3.22	ARIMA
MAPE (%)	33.40	2.18	ARIMA
MASE	5.65	0.27	ARIMA
AICc	635.72	512.06	ARIMA
Ljung-Box p	≈ 0.000	0.19	ARIMA

ARIMA wins on every metric without exception. The validation chart (Figure 7) visually confirms the ARIMA forecast tracks the actual installation values closely throughout the entire 2023–2024 window, particularly at the seasonal turning points where forecast precision has the greatest operational impact. TSLM's MASE of 5.65 places it well below the minimum deployment threshold, while ARIMA's MASE of 0.27 and AICc of 512.06 confirm it as the stronger and more parsimonious model.

Q12 — In the forecast plot, where are the biggest differences? Which model tracks actuals more closely?

- As seen in Figure 7, both models capture the general seasonal shape across the 2023–2024 validation window, however ARIMA tracks the actual installation values considerably more closely throughout.

The most notable divergence between the two models appears at the summer peaks (June–July 2023 and June–July 2024) and the winter troughs (January 2023 and January 2024), where the cyan TSLM line visibly overshoots and undershoots the actuals respectively. The orange ARIMA line, by contrast, remains tightly aligned with the black actual line at precisely these turning points where forecast accuracy carries the greatest operational significance for inventory and workforce planning.

The accuracy metrics confirm what the chart shows — ARIMA's RMSE of 3.84 against TSLM's 54.55 reflects a model that does not merely approximate the seasonal pattern but accurately tracks the magnitude and direction of demand at every point in the forecast horizon.

Q13 — TSLM has 13 parameters, ARIMA has 5. Does fewer parameters perform better here? What does this tell us about AICc?

- ARIMA's 5 parameters outperform TSLM's 13 on every single metric, demonstrating that more parameters do not guarantee better forecasts.

TSLM's 13 coefficients are entirely fixed estimated once from 2015–2022 data and locked in permanently. As SunBridge's demand accelerated and seasonal swings widened in 2023–2024, those frozen parameters had no capacity to adapt, resulting in a RMSE of 54.55 and a MASE of 5.65.

ARIMA's 5 parameters — two AR terms, two seasonal AR terms, and a drift constant — describe the dynamics of the series rather than its historical averages. Each parameter actively contributes to capturing momentum, seasonal relationships, and ongoing growth, delivering a RMSE of 3.84 and a MASE of 0.27 with a fraction of the complexity

This outcome reflects exactly what AICc is designed to identify. With an AICc of 512.06 against TSLM's 635.72, a gap of over 120 points, AICc correctly rewards ARIMA for achieving a far superior fit with fewer parameters. A model that adds complexity without proportionally improving forecast accuracy will always be penalised under AICc, and TSLM is a clear illustration of that principle.

Section 9 — Deployment Decision

Q14 — Which model would you deploy?

Recommendation: Deploy ARIMA(2,0,0)(2,1,0)[12] with drift.

All three diagnostic gates point to the same conclusion:

Check	TSLM	ARIMA	Result
(a) $MASE < 1$	5.65	0.27	ARIMA
(b) Ljung-Box $p > 0.05$	≈ 0.000	0.19	ARIMA
(c) $ME \approx 0$	+48.06	-1.63	ARIMA

(a) ARIMA's MASE of 0.27 confirms it is nearly four times more accurate than a seasonal naïve benchmark. TSLM's MASE of 5.65 means it fails the most basic accuracy standard required for any production forecasting model.

(b) ARIMA's Ljung-Box p-value of 0.19 confirms its residuals are white noise — no predictable structure has been left behind. TSLM's $p \approx 0.000$ confirms the opposite, making its forecast intervals statistically invalid.

(c) ARIMA's ME of -1.63 is operationally negligible. TSLM's chronic underprediction of 48 units every month would directly impact SunBridge's inventory ordering, crew scheduling, and revenue planning.

ARIMA passes all three gates; TSLM fails all three. The deployment decision is clear and fully supported by the evidence produced throughout this analysis.

Q15 — What are two limitations of your chosen model? What specific method would you try next?

- ARIMA(2,1,1)(0,1,1)[12] has proven to be the right model for SunBridge Solar — delivering a RMSE of 3.84, a near-zero bias of -1.63 units per month, and statistically clean residuals throughout the 2023–2024 validation period. That said, two honest limitations must be acknowledged.
 - **Limitation 1 — The model only learns from past installation data.**
ARIMA was built purely on SunBridge's own historical records. It has no knowledge of the outside factors that influence how many solar panels get installed each month — things like government tax credit programmes, changes in electricity prices, or the volume of new homes being built in the region. During our 2023–2024 validation period, market conditions were relatively stable, which helped ARIMA perform well. But as seen in Figure 1, demand grew sharply from 2020 onwards — a period that coincided with major federal solar incentives in the U.S. market. If a similar policy shift or market disruption were to occur in the future, the model would have no way of seeing it coming or adjusting for it.
 - **Limitation 2 — Peak season demand is growing faster than the model expects.**

As clearly shown in Figures 1 and 2 during the EDA, the gap between SunBridge's summer peaks and winter troughs has been getting wider every year. The model handles this reasonably well at today's volume levels, but as the business continues to grow, those summer peaks will become increasingly difficult to forecast accurately. Underestimating peak season demand — even by a small margin — directly translates to material shortages, understaffed crews, and missed installations at the busiest and most profitable time of the year.

What I recommend trying next : :ARIMAX.

The most practical and powerful next step is to upgrade to an ARIMAX model, which keeps everything that made ARIMA outperform TSLM — the dynamic error correction, the drift term, the seasonal structure — but adds the ability to include external business variables directly in the forecast. Based on the evidence from this analysis, two variables stand out as the most valuable starting points:

External Variable	Business Relevance
Federal solar tax credit rate	Directly explains the demand acceleration visible in Figure 1 from 2020 onwards that TSLM's +48 unit bias failed to capture
Regional housing permit volumes	New home construction leads solar installation demand by several months, giving the model early forward signal it currently lacks

If these external data sources are not yet available, ETS(M,A,M) is a strong alternative that is specifically built for series where seasonal swings grow proportionally with volume — addressing Limitation 2 immediately without requiring any additional data.

Citations

<https://otexts.com/fpp3/>

